

INFO6007

Project Management in IT

Week 08: Procurement Management Plan

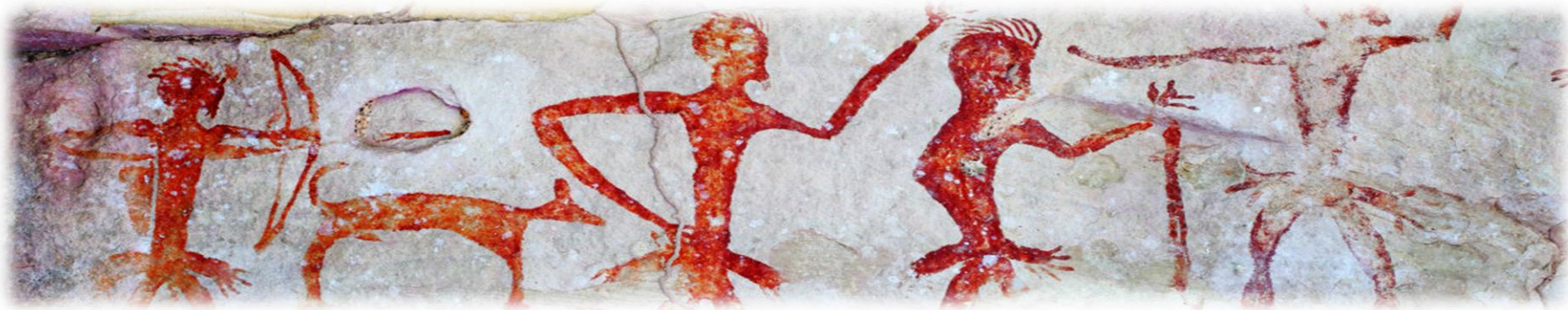
Semester 1, 2026



Acknowledgement of Country

I would like to acknowledge the Traditional Owners of Australia and recognise their continuing connection to land, water and culture. I am currently on the land of the Gadigal people of the Eora nation and pay my respects to their Elders, past, present and emerging.

I further acknowledge the Traditional Owners of the country on which you are on and pay respects to their Elders, past, present and future.



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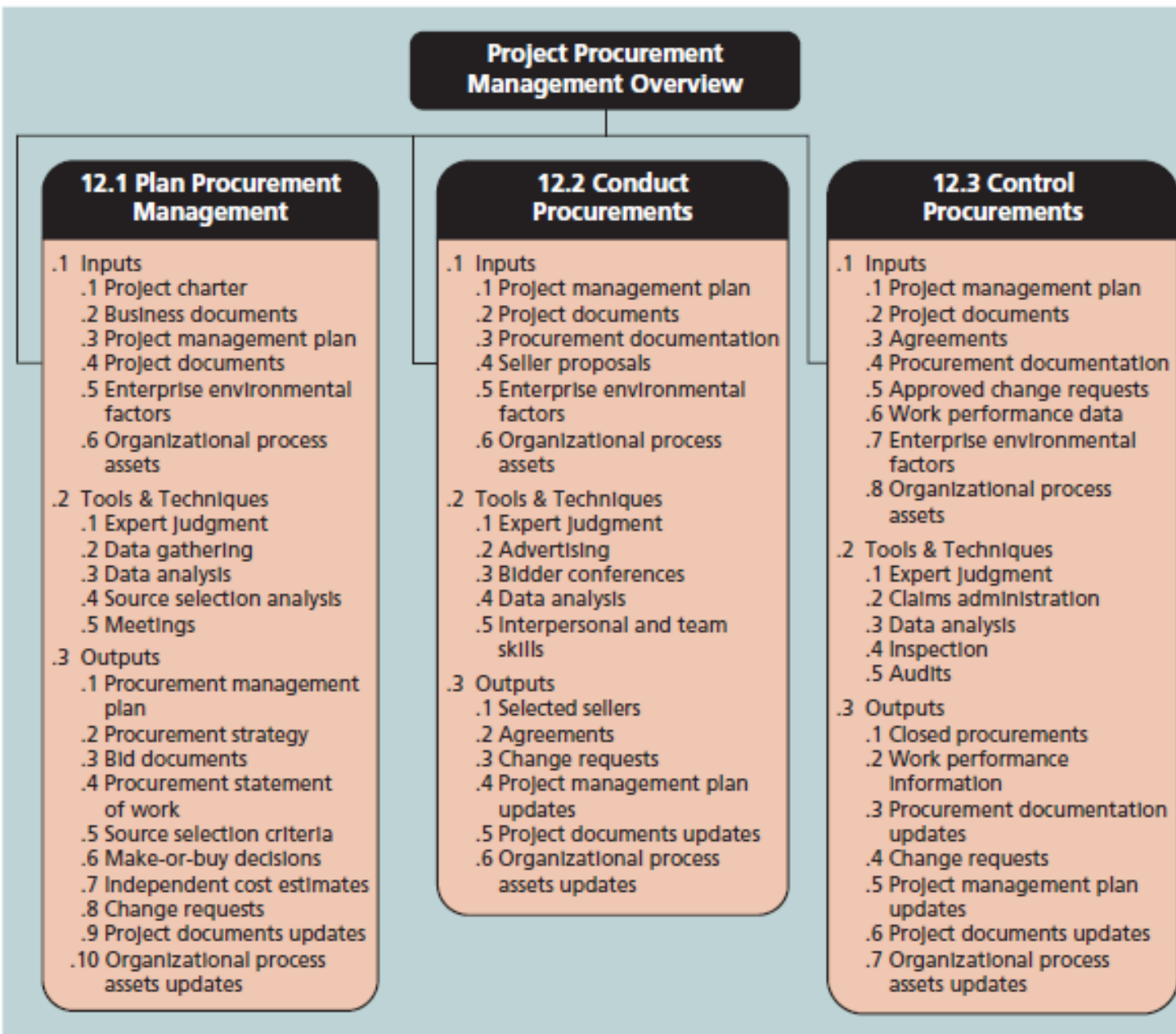
Agenda

- Introduction
- Project Procurement Management
- Components of Project Procurement Management
- Project Outsourcing

Project Procurement Management

Introduction

- Project Procurement Management is the process of acquiring goods, services, or results from outside the project team to meet project needs.
- In other words, it is about deciding what to buy, how to buy it, who to buy it from, and how to ensure the contract is successfully completed.
- Why does it matter?
 - Most IT projects rely on external parties for software, hardware, cloud services, consultants, or support.
 - Effective procurement ensures the right resources are delivered on time, within budget, and at the expected quality.
 - Poor procurement can lead to delays, cost overruns, and project failure.



3 steps of Project Procurement Management Plan

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FIGURE 12-1 Project procurement management overview

1. Plan Procurement Management

- Plan Procurement Management is the process of documenting project procurement decisions, specifying the approach, and identifying potential sellers.
- It answers:
 - What do we need to buy?
 - How will we buy it?
 - When will we buy it?
 - Who will manage it?

1. Plan Procurement Management

- The main purpose of Planning for procurement management is to:
 - Ensure that goods, services, or results are acquired in a structured, efficient, and cost-effective way.
 - Align procurement activities with project scope, schedule, budget, and risks.
 - Provide a roadmap for managing vendor relationships.
- An important technique to know is the Make-or-Buy Analysis

Make-or-Buy Analysis

- Make-or-Buy Analysis is the process of deciding whether a product, service, or result should be **produced internally (make)** or **purchased/outsourced externally (buy)**.
- It helps project managers determine the most cost-effective, time-efficient, and risk-appropriate option.
- A few factors to consider:

Cost

Expertise and skills

Time

Control and Confidentiality

Quality

Strategic Importance

Make-or-Buy Analysis

- Steps in Make-or-Buy Analysis
 - Identify the need (e.g., software component, IT service).
 - Estimate internal capability (resources, skills, costs, schedule).
 - Estimate external costs and constraints (vendor proposals, market research).
 - Compare options in terms of cost, risk, and benefits.
 - Document and justify decision in the Procurement Management Plan.

	Alternative 1 (Make Internally)	Alternative 2 (Buy from Outside)	Differential Amount	Alternative 1 Is
Variable costs				
Cost to buy from outside	\$ 00	\$ 700,000 ^a =	\$(700,000)	Lower
Direct materials	300,000 –	0 =	300,000	Higher
Direct labor	160,000 –	0 =	160,000	Higher
Manufacturing overhead	100,000 –	0 =	100,000	Higher
Fixed costs				
Factory equipment lease	110,000 –	110,000 =	0	
Factory building rent	290,000 –	290,000 =	0	
Production supervisors' salaries	140,000 –	90,000 ^b =	50,000	Higher
Total production costs	<u>\$1,100,000</u> –	<u>\$1,190,000</u> =	<u>\$ (90,000)</u>	Lower

Table shows example of make or buy analysis

https://saylordotorg.github.io/text_managerial-accounting/s11-02-make-or-buy-decisions.html

2. Conduct Procurements

- Conduct Procurements is the process of **obtaining seller responses, evaluating them, and awarding the contract**. It transforms procurement planning into action by interacting with the market to acquire the needed goods or services.
- Key activities:
 - Distribution of Procurement Documents
 - Vendor Conferences/Bidder meetings
 - Receive proposals/quotations
 - Evaluation of proposals
 - Negotiation with vendors
 - Selection and award of contract

3. Control Procurements

- Controlling Procurements is the process of **managing relationships with vendors, monitoring contract performance, and making changes as needed** to ensure that both the buyer and seller meet contractual obligations.
- The main objectives of this step is to:
 - Ensure the vendor delivers as per contract scope, schedule, cost, and quality.
 - Identify and address issues (delays, cost overruns, non-compliance).
 - Manage contract changes and disputes fairly.
 - Maintain proper communication and documentation.

Tips for Controlling Procurements

- Monitor performance regularly
- Keep communication transparent
- Manage change formally
- Verify before payment
- Build strong relationships
- Manage risks proactively
- Ensure compliance
- Document everything!

Components of Procurement Management

Introduction

- Procurement Management is one of the **PMBOK knowledge areas** and consists of several structured components that guide how a project acquires goods and services from outside the organization.
- The following will introduce you to various documentations required for Project Procurement Planning.

Procurement Statement of Work

- A **Statement of Work (SOW)** is a detailed description of the products, services, or results to be delivered under a contract.
- It sets the **scope, deliverables, timelines, and standards** so that both the buyer and seller understand exactly what is expected.
- Think of it as the **"what and how" document** for procurement.
- It's main purpose is to:
 - Provides clarity to vendors on requirements.
 - Forms the basis for preparing proposals (RFP responses).
 - Reduces misunderstandings and scope creep.
 - Acts as a legal reference in case of disputes.

Statement of Work (SOW)

- I. **Scope of Work:** Describe the work to be done in detail. Specify the hardware and software involved and the exact nature of the work.
- II. **Location of Work:** Describe where the work must be performed. Specify the location of hardware and software and where the people must perform the work.
- III. **Period of Performance:** Specify when the work is expected to start and end, working hours, number of hours that can be billed per week, where the work must be performed, and related schedule information.
- IV. **Deliverables Schedule:** List specific deliverables, describe them in detail, and specify when they are due.
- V. **Applicable Standards:** Specify any company or industry-specific standards that are relevant to performing the work.
- VI. **Acceptance Criteria:** Describe how the buyer organization will determine if the work is acceptable.
- VII. **Special Requirements:** Specify any special requirements such as hardware or software certifications, minimum degree or experience level of personnel, travel requirements, and so on.

Procurement Statement of Work - Example

Request for Proposal

- A **Request for Proposal (RFP)** is a formal document issued by a buyer (the project team or organization) to invite potential sellers/vendors to submit **proposals** on how they would meet the project's requirements.
- It provides enough detail about the project so vendors can propose **solutions, costs, schedules, and approaches**.
- The main purpose of this document is to:
 - To **communicate requirements** clearly to potential vendors.
 - To **encourage competition** and obtain the best value.
 - To **evaluate proposals** based on technical, financial, and qualitative factors.
 - To **ensure transparency and fairness** in vendor selection.

Request for Proposal

- The key components of an RFP are:
 - Introduction and Background
 - Scope of Work [References the Statement of Work]
 - Requirements
 - Proposal Guidelines
 - Evaluation criteria
 - Timeline
 - Budget (optional) [May be provided or left open for vendor proposals]
 - Terms and conditions [Contract types, payment terms and legal conditions]

Request for Proposal

- The RFP process is as follows:
 - Buyer prepares RFP.
 - RFP is published/shared with vendors.
 - Vendors prepare proposals (solution, cost, schedule).
 - Buyer evaluates proposals using selection criteria.
 - Vendor is selected and contract negotiations begin.

Request for Quotation

- A **Request for Quotation (RFQ)** is a formal procurement document issued by a buyer to solicit price quotations from suppliers for well-defined goods or services.
- Unlike an RFP (Request for Proposal), which seeks detailed solutions, an RFQ is primarily focused on **price, delivery terms, and payment conditions** when the requirements are clear and standardized.
- The main purpose of the RFQ is:
 - To obtain **competitive pricing** for specific items or services.
 - To identify the **most cost-effective supplier** that meets technical and quality requirements.
 - To establish a **baseline for vendor comparison** when scope and specifications are already fixed.

Request for Quotation

- Key features of RFQ:
 - **Detailed Specifications:** The buyer provides clear technical requirements (e.g., model numbers, sizes, quantities).
 - **Price-Centric Evaluation:** Focuses mainly on costs, discounts, and payment terms.
 - **Standardized Products/Services:** Used when items are common, commoditized, and easily comparable.
 - **Short Decision Cycle:** Faster procurement process compared to RFP.
- When to use an RFQ?
 - Buying commodities (e.g., office supplies, raw materials).
 - Procuring standardized IT hardware/software licenses.
 - When requirements are clear and do not need innovation or customization.
 - For comparisons across multiple suppliers mainly on cost and delivery.

RFP vs RFQ

Aspect	Request for Quotation	Request for Proposal
Definition	A document to solicit price quotations from suppliers for well-defined goods or services	A document issued by a buyer to invite potential sellers/vendors to submit proposals on how they would meet the project's requirements.
Focus	Price & delivery	Technical solution + price
Requirements	Well defined, standardised	Complex, needs innovation
Evaluation criteria	Cost driven	Quality, approach, and capability
When to use?	When you need price quotes for well-defined items.	When you want solutions and vendor creativity.
Typical use case	Commodities, shared items	IT projects, consulting, custom systems

Types of Contracts

- When a project team engages with a vendor/supplier, the agreement is formalized through a **contract**. The type of contract defines:
 - How payment is structured
 - How risk is shared between buyer and seller
 - How performance is measured
- There are mainly 3 types of contracts:
 - Fixed price contracts
 - Cost reimbursable contracts
 - Time & materials contracts

Further types of contracts

- Out of the 3 types of contracts, 2 are further divided into sub-types:
 - Fixed-Price contracts
 - **Firm Fixed Price (FFP)**: Price is final, no adjustments.
 - **Fixed Price Incentive Fee (FPIF)**: Price fixed, but with performance incentives.
 - **Fixed Price with Economic Price Adjustment (FP-EPA)**: Price fixed, but adjusted for inflation/market changes.
 - Cost-Reimbursable contracts
 - **Cost Plus Fixed Fee (CPFF)**: Seller reimbursed + fixed fee.
 - **Cost Plus Incentive Fee (CPIF)**: Seller reimbursed + incentive based on performance.
 - **Cost Plus Award Fee (CPAF)**: Seller reimbursed + award based on buyer's satisfaction.

Further types of contracts vs risk

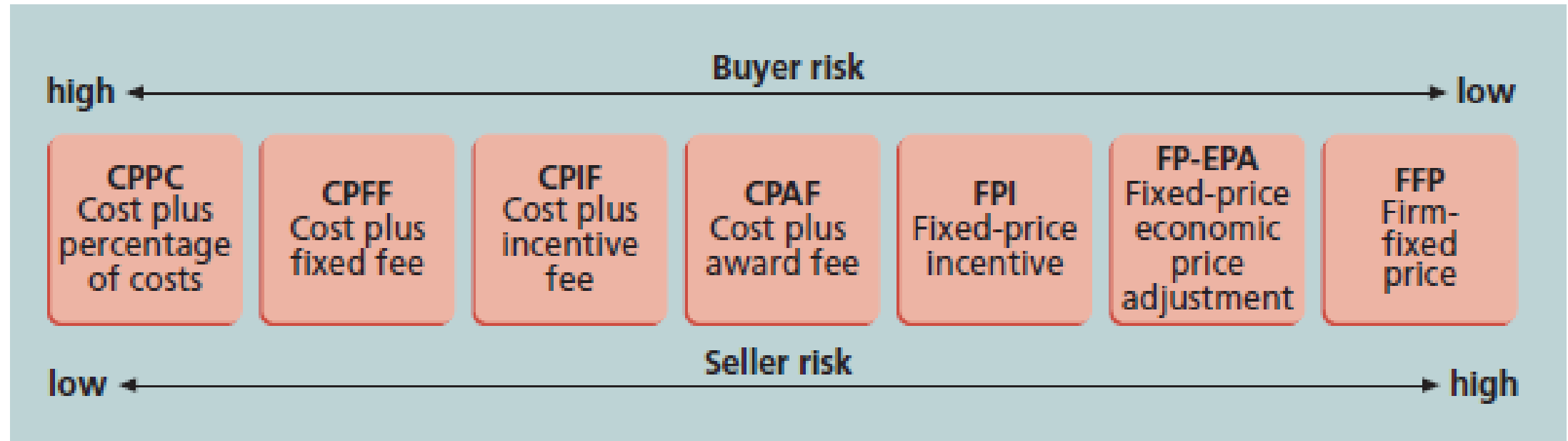


FIGURE 12-2 Contract types versus risk

Types of Contracts

Aspect	Fixed-Price contracts	Cost-Reimbursable contracts	Time and Material contracts
Definition	Price is agreed upon in advance, regardless of actual costs.	Buyer reimburses seller for actual costs, plus a fee/profit.	Hybrid of fixed and cost-reimbursable. Buyer pays per hour/day plus materials.
Risk on Buyer	Low	High	Medium
Risk on Seller	High	Low	Medium
Advantages	Predictable cost for buyer, risk on seller.	Flexibility, good for R&D or uncertain projects.	Simple to administer, flexible scope.
Disadvantages	Seller may cut corners if costs rise; change requests are expensive.	Higher cost risk for buyer, less incentive for seller to control costs.	Risk of cost overruns if not monitored.
Best use case	Clear scope, predictable deliverables	R&D, uncertain requirements	Staff augmentation, flexible scope

Vendor Evaluation Criteria

- This helps ensure that the organization selects the right supplier not just based on price, but on overall **value, quality, and risk management**.
- A few possible selection criteria are:
 - **Cost and pricing:** Total cost of ownership, Transparency of pricing, and Flexibility in payment terms.
 - **Quality of goods/services:** Compliance with specifications and standards.
 - **Technical capability:** Availability of skilled personnel and technical know-how.
 - **Customer service and support:** Responsiveness to queries, complaints, and service issues.

Vendor Evaluation Methods

- These are the structured ways organizations assess and compare vendors to make procurement decisions.
- There are 3 commonly used methods:
 - Weighted Scoring Model
 - **Best for:** Complex procurements where multiple factors matter, not just price.
 - Checklist
 - **Best for:** When compliance to minimum standards (e.g., certifications, licenses) is critical.
 - Vendor Scorecard
 - **Best for:** Long-term supplier relationships.

Vendor Evaluation: Weighted Scoring

Potential Suppliers Evaluation

Criteria	Supplier A			Supplier B	
	Weight	Score	Weight x Score A	Score	Weight x Score B
Integrity	0.20	1	0.20	1	0.20
Industry Expertise	0.35	3	1.05	5	1.75
Experience and Qualification	0.20	2	0.40	4	0.8
Financial and Managerial Strength	0.25	1	0.25	0	0
Total		(0 = Low, 5 = High)	1.90	(0 = Low, 5 = High)	2.75

Select Supplier B
1.90 < 2.75

Source: Finding, Evaluating and Selecting Suppliers – Procurement in the Supply Chain World” (Pressbooks, Ontario)

[Finding, Evaluating and Selecting Suppliers – Procurement in the Supply Chain World](#)

Vendor Evaluation: Checklist

Criteria	Supplier A	Supplier B	Supplier C
Legal & Compliance	✓	✓	✓
Certifications	ISO 9001	ISO 22000	ISO 9001, HACCP
Financial Stability	Stable	At risk	Stable
Capacity & Capability	Yes	No	Yes
Quality Standards	Yes	Yes	Yes
Delivery Performance	95%	80%	92%
Cost Competitiveness	Competitive	High	Competitive
Customer References	Positive	Mixed	Positive
Sustainability & Ethics	Yes	No	Yes
Risk Factors	Low	Medium	Low
Overall Recommendation	☑	☐	☑

Prepared with assistance from OpenAI's ChatGPT

Vendor Evaluation: Vendor Scorecard

- Purpose:** Ongoing performance monitoring of suppliers already engaged.
- Focus:** Tracks key performance indicators (KPIs) such as delivery timeliness, quality, cost, customer service, and compliance.
- Nature:** Retrospective and continuous - used after awarding the contract to evaluate how well the supplier is performing.
- Format:** Typically a dashboard or report with scores, ratings, or traffic lights (green/yellow/red).
- Example:** Monthly supplier performance review showing *on-time delivery 92% (green)*, *defect rate 4% (yellow)*.

Menti

- *USyd is tendering for a new catering supplier for campus events. The main requirement is that the supplier must comply with food safety certifications, have valid insurance, and meet health and hygiene regulations. Price and menu variety are secondary considerations.*
- **Question:**
Which supplier evaluation method is most suitable in this case?
- <https://www.menti.com/alyxjzezogo2>



Procurement Schedule

- A **Procurement Schedule** is a timeline that outlines the key activities, milestones, and deadlines for managing procurement activities in a project. It aligns procurement processes with the overall **project schedule** to avoid delays.
- Key elements of a project procurement schedule:
 - Procurement planning timeline
 - Document preparation
 - Vendor selection timeline
 - Contract award milestone
 - Delivery and performance periods
 - Monitoring and controlling timeline
 - Procurement closure

Risk Management in Procurement

- Procurement risk management is the process of **identifying, assessing, and mitigating risks** that arise when acquiring goods, services, or works from external vendors.
- It ensures that vendor relationships, contracts, and deliverables do not jeopardize project success.
- Common risks in procurement:
 - **Cost:** Price fluctuations, hidden costs, penalties, or cost overruns.
 - **Schedule:** Delays in delivery, long lead times, logistics and transportation issues.
 - **Quality:** Products/Services do not meet specifications, vendor cutting corners.
 - **Compliance and legal:** Breach of contract terms, IP disputes, etc.
 - **Operational:** Miscommunication between vendor and buyer, poor contracts, etc.

Steps in Procurement Risk Management

- **Risk identification**
 - Brainstorm potential risks with project and procurement teams.
 - Review vendor history, market conditions, and past lessons learned.
- **Risk assessment**
 - Analyze risks based on likelihood and impact.
 - Use a risk matrix to prioritize (e.g., high/medium/low).
- **Risk mitigation strategies**
 - Cost risks: Use fixed-price contracts, hedging, or multi-year agreements.
 - Schedule risks: Include penalties for delays (liquidated damages).
 - Quality risks: Define strict acceptance criteria in the Statement of Work (SOW).

Steps in Procurement Risk Management

- Risk monitoring
 - Track vendor performance with KPIs.
 - Regular audits, inspections, and progress reports.
- Risk response and contingency
 - Have contingency plans (e.g., secondary supplier).
 - Escalation procedures for resolving disputes quickly.

Project Outsourcing

Introduction

- Project outsourcing is the practice of **hiring an external organization or vendor** to perform specific project activities, tasks, or deliverables instead of executing them in-house.
- It is often used to leverage **specialized expertise, reduce costs, or accelerate timelines**.
- Why to outsource projects?
 - **Cost Efficiency:** Reduce labor, infrastructure, and operational costs.
 - **Access to Expertise:** Gain specialized skills not available internally (e.g., cloud architecture, AI development).
 - **Focus on Core Activities:** Let the organization focus on strategic activities while vendors handle support or technical work.
 - **Risk Sharing:** Vendors may assume certain risks related to delivery, technology, or compliance.

Types of Outsourcing in IT Projects

- Complete Project Outsourcing
 - The vendor manages the **entire project** from initiation to closure.
 - Example: Outsourcing the development of a mobile app from design to deployment.
- Partial Project Outsourcing
 - Only **specific components** or tasks are outsourced.
 - Example: A company develops software in-house but outsources testing or UI/UX design.
- Business Process Outsourcing (BPO)
 - Outsourcing of **non-core business processes** related to the project.
 - Example: Customer support, payroll processing, or IT helpdesk services.
- IT/Technical Outsourcing
 - Hiring specialized vendors for **software development, cloud infrastructure, or cybersecurity**.
 - Example: Cloud migration, AI model development, etc.

Outsourcing Models

Model	Description	Example
Onshore	Vendor in same country	Local software development company
Nearshore	Vendor in nearby country/time zone	Indonesian vendor for Australian company
Offshore	Vendor in distant country	India or Vietnam outsourcing for cost savings
Hybrid	Mix of internal and external resources	Core strategy in-house, coding offshore
Multisource	A mix of the different models of sourcing	
Insourcing	Opposite of outsourcing. Movement of business process within the company but to another internal entity.	A company develops its mobile app using its internal software development team instead of hiring an external vendor.

Benefits of Outsourcing IT Projects

- Cost reduction and access to global talent.
 - Faster project completion due to experienced vendors.
 - Reduced burden on internal teams.
 - Flexibility in resource management.
-
- Access skills and technologies unavailable within the company.
 - Reduce both fixed and recurrent costs.
 - Allows client organisation to focus on its core business.

Disadvantages of Project Outsourcing

- Communication issues across geographies/time zones.
- Quality control and adherence to standards.
- Confidentiality and data security concerns.
- Vendor dependency and lack of internal skill development.
- Hidden costs in coordination or contract management.

Governance of Outsourcing

- Outsourcing governance is the **framework of policies, processes, roles, and oversight** that ensures outsourced activities align with organizational objectives, deliver value, and manage risks effectively.
- It ensures accountability, performance monitoring, and compliance in vendor relationships.
- Key components of Governance of outsourcing:
 - Strategic alignment: Ensure outsourced services align with **business goals** and project objectives.
 - Contract management: Oversight of **contract adherence**, including deliverables, milestones, penalties, and incentives.
 - Roles and responsibilities: Define **who in the organization manages the vendor** (e.g., Vendor Manager, Project Manager) and specify vendor responsibilities, deliverables, and escalation points.

Integration of Procurement and Outsourcing in IT Projects

- Why is this integration important?
 - **Alignment with Project Objectives:** Ensures procured or outsourced components meet technical and business requirements.
 - **Risk Reduction:** Minimizes schedule delays, cost overruns, and quality issues.
 - **Efficient Resource Utilization:** Combines internal and external resources effectively.
 - **Better Vendor Management:** Provides clear accountability, performance monitoring, and contract compliance.
 - **Strategic Decision-Making:** Helps decide what to build in-house vs outsource, and how to procure critical components.

Best practices for integration

- **Start Early:** Include procurement and outsourcing planning in the project initiation phase.
- **Align Contracts with Project Milestones:** Ensure payments and deliverables match the project schedule.
- **Use Integrated Tools:** Track both internal and external tasks in the same project management system.
- **Continuous Communication:** Maintain transparency between internal teams and vendors.
- **Mitigate Risks Proactively:** Identify dependencies and maintain backup plans (e.g., alternate vendors, contingency inventory).

References

- Schwalbe, K., 2018. Project Risk Management. In: *Information Technology Project Management*. Boston: Cengage, pp. 505-538.

For You!

End of Lecture Questions

- Define **project procurement management** and explain its importance in IT projects.
- Differentiate between **insourcing** and **outsourcing** in project management.
- What is a **Make-or-Buy Analysis**, and why is it critical in procurement planning?
- List and briefly describe the **types of contracts** used in outsourcing.

Case Study

- TechSolutions Pty Ltd., a mid-sized manufacturing company in Sydney, decides to implement a **cloud-based ERP system** to improve operations, inventory management, and financial reporting.
- The internal IT team lacks expertise in ERP customization and cloud deployment. The company plans to outsource key components while procuring necessary hardware and software licenses.

- Project details

Internal Team: Project management, system integration, change management.

Outsourced Tasks: ERP software customization, cloud deployment, data migration, cybersecurity audits.

Procured Items: Servers for hybrid cloud backup, software licenses, network infrastructure.

Vendors Selected: ERP Development Vendor (Offshore), Cloud Hosting Provider (Onshore), Cybersecurity Auditing Firm (Nearshore)

- Timeline

Project initiation: Week 1

Vendor selection: Weeks 2–4

Contract award: Week 5

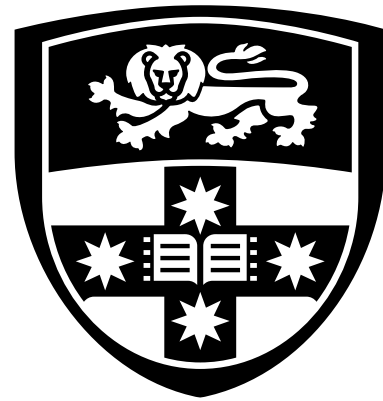
ERP deployment: Weeks 6–16

Testing & acceptance: Weeks 17–18

Project closure: Week 20

Case-study based questions

- Identify which components of the project are **insourced** and which are **outsourced**.
- Explain the **outsourcing models** used for each vendor.
- Discuss the importance of **procurement planning** in this project.
- Identify **risks** associated with each vendor and suggest **risk mitigation strategies**.
- Describe how the project manager can **control procurements** to ensure vendor deliverables are on time and meet quality standards.
- If the offshore ERP vendor fails to deliver on time, how could **multisourcing** or alternative strategies have reduced project risk?
- Discuss how this project demonstrates the importance of **controlling procurement risks** in IT projects.



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